Table 1:Change log

Issue number: 1			
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Version	**Date**	**Comments**	**Authors**
1	2023-07-07	Creation of document	Chiara Monforte and Callum Rollo
2	2023-07-10	Added info on new missions: SEA044_M83 and SEA063_M62	Chiara Monforte
3	2023-07-27	Added info on new missions: SEA067_M46; SEA069_M15; SEA077_M24 and SEA078_M14	Chiara Monforte
4	2023-08-04	Added info on new mission: SEA044_M84	Chiara Monforte
5	2023-08-14	Added info on new mission: SEA063_M63	Chiara Monforte
6	2023-09-21	Added info on new mission: SEA056_M64; SEA079_M14; SEA044_M85; SEA044_M86 and SEA076_M19	Chiara Monforte
7	2023-09-27	Added info on new mission: SEA063_M65	Chiara Monforte
8	2023-10-06	Added info on new mission: SEA077_M25 and SEA078_M15	Chiara Monforte
9	2023-10-10	Added info on new mission: SEA069_M17	Chiara Monforte
10	2023-10-18	Added info on new mission: SEA056_M66	Chiara Monforte
11	2023-10-23	Added info on new mission: SEA069_M19	Chiara Monforte
12	2023-11-13	Added info on new mission: SEA063_M67 and SEA079_M16	Chiara Monforte
13	2023-11-20	Added info on new mission: SEA069_M21; SEA066_M52 and SEA076_M21	Chiara Monforte
14	2023-11-29	Added info on new mission: SEA067_M51	Chiara Monforte
15	2023-12-02	Added info on new mission: SEA069_M22	Chiara Monforte
16	2023-12-07	Added info on new mission: SEA077_M28 and SEA078_M19	Chiara Monforte

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Table 1:Change log (Continued)

17	2023-12-15	Added info on new mission: SEA056_M68 and SEA063_M69	Chiara Monforte
18	2023-12-27	Added info on new mission: SEA069_M23	Chiara Monforte
19	2024-01-08	Added info on new mission: SEA076_M22 and SEA079_M18	Chiara Monforte
20	2024-01-10	Added info on new mission: SEA077_M29 and SEA067_M53	Chiara Monforte
21	2024-01-18	Added info on new mission: SEA044_M88	Chiara Monforte
22	2024-01-30	Added info on new mission: SEA068_M30	Chiara Monforte
23	2024-02-08	Added info on new mission: SEA078_M22; SEA056_M71; SEA076_M24 and SEA069_M25	Chiara Monforte
24	2024-02-20	Added info on new mission: SEA063_M71	Chiara Monforte
25	2024-03-08	Added info on new mission: SEA044_M89, SEA77_M31 and SEA079_M20	Chiara Monforte
26	2024-03-13	Added info on new mission: SEA067_M55	Chiara Monforte
27	2024-03-20	Added info on new mission: SEA078_M24	Chiara Monforte
28	2024-03-25	Added info on new mission: SEA069_M26 and SEA063_M72	Chiara Monforte

1 Introduction

A fault has affected dissolved oxygen concentration values on all gliders with the AROD RINKO dissolved oxygen sensor deployed from 2022-12-01. In the affected datasets, oxygen concentration is around 100 µmol m^{-3} less than expected. This offset decreased during the mission at a non linear rate. All affected datasets have been flagged with the variable oxygen_concentration_qc . We are working to fix the affected sensors.

The table below (Table 2) shows an updated list of all the mission deployed after 2022-12-01 with an AROD RINKO sensor, color-coded by severity. Flag column corresponds to data quality: suspect (3), fail (4), and good (1).

Table 2: Info summary for all the missions deployed after 2022-12-01 with an AROD RINKO sensor. The different basins mentioned in the 'Location' column, follow the division made by HELCOM which is shown in A.1

Glider	Mission	Location	Mission Start date	Sensor Serial	Issue description	Flag
SEA044	72	Skagerrak, Kattegat	2023-02-06	70	B	4
	75	Skagerrak, Kattegat	2023-03-18	70	C	3
	77	Skagerrak, Kattegat	2023-04-14	70	D	4
	78	Skagerrak, Kattegat	2023-05-04	70	D	4
	79	Skagerrak, Kattegat	2023-05-31	70	D	4
	81	Skagerrak, Kattegat	2023-06-16	70	D	4
	83	Skagerrak, Kattegat	2023-07-10	68	E	4
	84	Skagerrak, Kattegat	2023-08-04	68	E	4
	85	Skagerrak, Kattegat	2023-08-23	68	E	4
	86	Skagerrak, Kattegat	2023-09-14	68	E	4
	88	Skagerrak, Kattegat	2024-01-18	58	A	1
	89	Skagerrak, Kattegat	2024-03-04	58	A	1
SEA045	73	Bornholm Basin	2022-12-16	39	C	3
	74	Bornholm Basin	2023-02-02	39	B	4
	75	Bornholm Basin	2023-03-16	39	B	4
	79	Bornholm Basin	2023-05-30	58	E	4
SEA055	52	Skagerrak, Kattegat	2022-12-01	58	C	3
	53	Skagerrak, Kattegat	2022-12-21	58	C	3
	54	Skagerrak, Kattegat	2023-01-20	58	C	3
	56	Skagerrak, Kattegat	2023-02-27	58	B	4
SEA056	56	Bornholm Basin	2022-12-09	50	C	3
	57	Bornholm Basin	2022-12-17	50	C	3

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Table 2: Info summary for all the missions deployed after 2022-12-01 with an AROD RINKO sensor. The different basins mentioned in the 'Location' column, follow the division made by HELCOM which is shown in A.1 (Continued)

	62	Bornholm Basin	2023-05-09	69	D	4
	64	Bornholm Basin	2023-09-06	39	E	4
	66	Bornholm Basin	2023-10-18	39	E	4
	68	Åland Sea	2023-12-14	67	A	1
	71	Western Gotland Basin	2024-02-06	39	E	4
SEA063	55	Bornholm Basin	2023-03-01	71	C	3
	60	Bornholm Basin	2023-04-22	39	D	4
	62	Bornholm Basin	2023-07-08	50	E	4
	63	Bornholm Basin	2023-08-11	50	E	4
	65	Bornholm Basin	2023-09-27	50	E	4
	67	Bornholm Basin	2023-11-13	50	E	4
	69	Bornholm Basin	2023-12-15	69	E	4
	71	Bornholm Basin	2024-02-20	40	A	1
	72	Bornholm Basin	2024-03-25	40	A	1
SEA066	52	Western Gotland Basin	2023-11-18	67	A	1
SEA067	41	Bornholm Basin	2023-04-04	39	D	4
	42	Bornholm Basin	2023-05-09	67	D	4
	44	Bornholm Basin	2023-06-18	58	D	4
	46	Bornholm Basin	2023-07-26	39	E	4
	51	Bornholm Basin	2023-11-29	39	E	4
	53	Bornholm Basin	2024-01-09	39	E	4
	55	Bornholm Basin	2024-03-13	41	A	1
SEA068	30	Bornholm Basin	2024-01-30	67	A	1
SEA069	15	Bornholm Basin	2023-07-26	50	E	4
	17	Skagerrak, Kattegat	2023-10-10	68	E	4
	19	Skagerrak, Kattegat	2023-10-23	68	E	4
	21	Skagerrak, Kattegat	2023-11-17	66	E	4
	22	Skagerrak, Kattegat	2023-12-02	66	E	4
	23	Skagerrak, Kattegat	2023-12-27	66	E	4
	25	Skagerrak, Kattegat	2024-02-08	66	E	4
	26	Skagerrak, Kattegat	2024-03-25	66	E	4
SEA076	13	Northern Baltic Proper, Eastern Gotland Basin	2023-01-13	41	C	3

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Table 2: Info summary for all the missions deployed after 2022-12-01 with an AROD RINKO sensor. The different basins mentioned in the 'Location' column, follow the division made by HELCOM which is shown in A.1 (Continued)

	16	Eastern Gotland Basin, Northern Baltic Proper	2023-04-11	66	D	4
	17	Western Gotland Basin	2023-06-20	66	E	4
	19	Western Gotland Basin	2023-09-06	41	E	4
	21	Åland Sea	2023-11-17	70	A	1
	22	Western Gotland Basin	2024-01-07	70	A	1
	24	Åland Sea	2024-02-07	70	A	1
SEA077	21	Eastern Gotland Basin	2023-03-16	66	B	4
	22	Eastern Gotland Basin, Northern Baltic Proper	2023-05-16	41	D	4
	24	Eastern Gotland Basin, Northern Baltic Proper	2023-07-28	69	E	4
	25	Eastern Gotland Basin, Northern Baltic Proper	2023-10-06	69	E	4
	28	Eastern Gotland Basin, Northern Baltic Proper	2023-12-06	40	A	1
	29	Åland Sea	2024-01-08	40	A	1
	31	Western Gotland Basin	2024-03-08	68	A	1
SEA078	11	Northern Baltic Proper, Eastern Gotland Basin	2023-03-16	68	B	4
	12	Eastern Gotland Basin	2023-05-16	71	D	4
	14	Western Gotland Basin	2023-07-28	66	E	4
	15	Western Gotland Basin	2023-10-06	66	E	4
	19	Western Gotland Basin	2023-12-06	58	A	1
	22	Northern Baltic Proper, Eastern Gotland Basin	2024-02-06	69	E	4
	24	Åland Sea	2024-03-20	71	A	1
SEA079	9	Northern Baltic Proper, Eastern Gotland Basin	2023-02-14	69	B	4
	11	Eastern Gotland Basin	2023-04-11	68	D	4
	12	Eastern Gotland Basin, Northern Baltic Proper	2023-06-20	69	E	4
	14	Eastern Gotland Basin, Northern Baltic Proper	2023-09-06	71	E	4

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Table 2: Info summary for all the missions deployed after 2022-12-01 with an AROD RINKO sensor. The different basins mentioned in the 'Location' column, follow the division made by HELCOM which is shown in A.1 (Continued)

	16	Eastern Gotland Basin, Northern Baltic Proper	2023-11-11	81	A	1
	18	Eastern Gotland Basin, Northern Baltic Proper	2024-01-07	81	A	1
	20	Eastern Gotland Basin, Northern Baltic Proper	2024-03-08	81	A	1

Legend		Flag
A	Data is good	1
B	Deterioration of the foil has caused uncorrectable errors in the dissolved oxygen concentration. Values are observed to be substantially lower than the reference CTD cast at deployment. There is an apparent non-linear temporal increase in dissolved oxygen concentration throughout the mission.	4
C	Dissolved oxygen concentration values may be inaccurate. Deterioration of the foil has caused uncorrectable errors and non-linear temporal drift in other missions with this sensor during this timeframe. We have not observed issues in this particular mission, but we recommend caution when using this data.	3
D	The absolute values of the oxygen data can be unreliable. The relative variability is confirmed to be valid and no temporal drift exist has been observed. Absolute values may be inaccurate and lower than reference data. Re-calibration of the sensor is required.	4
E	Following issues with the oxygen sensor, a 2-point calibration has been performed (the exact date can be found in the mission metadata). This 2-point calibration has been applied to the data. The values at 0 and 100 % saturation are accurate. Absolute values between these two extremes may be inaccurate. The relative variability is confirmed to be valid and no temporal drift occurs. During tests, values outside the manufacturer acceptance range have been observed when compared to the output from a reference sensor. This issue is particularly evident around 100µmol/l.	4

2 Examples

Few example missions from the Gotland Basin are shown below (Figure 1). Oxygen saturation over depth is presented and data is colored by time (in days since deployment). Some missions exhibit an evident temporal drift (SEA079_M9; SEA077_M21; SEA078_M11) with saturation values increasing from around 50% to just below 100% in about 25 days.

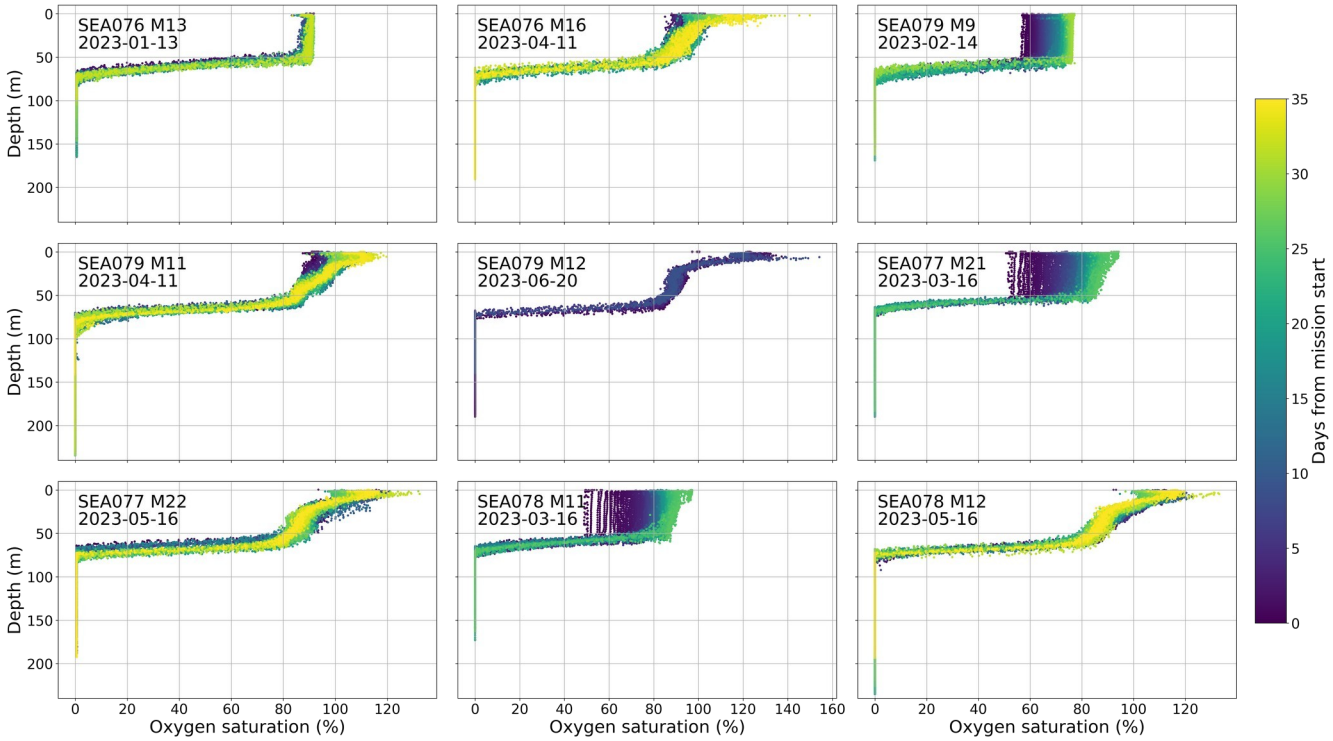


Figure 1: Oxygen saturation plotted over depth and colored by time (in days since deployment) for 9 missions in the Gotland Basin. Glider serial and mission number as well as the deployment date are noted on the top left of each subplot.

A Appendix



Figure A.1: Map of the Baltic Sea showing the 17 sub-basins(Map taken from http://stateofthebalticsea.helcom.fi/in-brief/our-baltic-sea/)